

SMART WATER TANK MONITORING SYSTEM USING IoT

Project Report

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ABSTRACT

Water is an essential resource, and efficient water management is important in residential buildings, schools, industries, and public facilities. Manual monitoring of water tank levels can lead to water overflow, water wastage, and unnecessary effort.

The **Smart Water Tank Monitoring System using IoT** is designed to monitor the water level of a tank automatically using an **HC-SR04 ultrasonic sensor** and an **Arduino Uno**. The ultrasonic sensor measures the distance between the sensor and the water surface. The Arduino processes this distance and calculates the percentage of water present in the tank.

When the water level falls below **20%**, the system identifies it as a low-water condition and activates a buzzer. When the water level reaches **90% or above**, the system identifies the tank as full and activates the buzzer. A **SIM800L GSM module** is used to send SMS alerts to the user.

This system provides real-time monitoring, reduces water wastage, minimizes manual supervision, and helps users take timely action when the tank is empty or full.

1. INTRODUCTION

Water tanks are commonly used to store water in houses, apartments, schools, hospitals, industries, and commercial buildings. Traditionally, users have to manually check the water level of the tank. This can be inconvenient and may result in overflow when the motor is not switched off at the appropriate time.

The Smart Water Tank Monitoring System provides an automatic solution for monitoring the water level. An ultrasonic sensor is installed at the top of the tank to measure the distance from the sensor to the water surface.

The Arduino Uno receives the sensor data and calculates the water level as a percentage. Based on predefined limits, the system identifies the tank condition as **LOW, NORMAL, or FULL**.

The SIM800L GSM module provides remote notification by sending SMS messages to the user's mobile phone. Therefore, the user can know the condition of the tank without physically checking it.

2. PROBLEM STATEMENT

Manual monitoring of water tanks has several problems:

- Water tanks may overflow when the motor is not switched off.
- Water may become insufficient when the tank level is not checked regularly.
- Manual monitoring requires time and effort.
- Water wastage can occur due to overflow.
- Users may not be near the tank when the water level becomes critical.
- There is a need for a simple and low-cost automatic monitoring system.

The proposed system addresses these problems by continuously monitoring the water level and providing local and remote alerts.

3. OBJECTIVES

The main objectives of the project are:

1. To monitor the water level of a tank automatically.
2. To measure the distance between the sensor and water surface using an ultrasonic sensor.

3. To calculate the water level as a percentage.
 4. To provide a buzzer alert when the water level is low or full.
 5. To send SMS notifications using the SIM800L GSM module.
 6. To reduce water wastage caused by tank overflow.
 7. To reduce the need for manual water-level monitoring.
 8. To develop a simple, affordable, and practical IoT-based monitoring system.
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4. COMPONENTS REQUIRED

S.N o	Component	Purpose
1	Arduino Uno	Main controller
2	HC-SR04 Ultrasonic Sensor	Measures distance to water surface
3	SIM800L GSM Module	Sends SMS alerts
4	Buzzer	Provides audible alerts
5	Jumper Wires	Electrical connections
6	Breadboard	Prototyping and connections
7	Power Supply	Provides power to the circuit
8	Water Tank	System monitoring environment

5. HARDWARE DESCRIPTION

5.1 Arduino Uno

The Arduino Uno is the main controller of the system. It receives distance measurements from the HC-SR04 sensor, calculates the water level, checks the predefined thresholds, controls the buzzer, and communicates with the SIM800L GSM module.

The Arduino Uno is based on the ATmega328P microcontroller and provides sufficient digital input/output pins for this project.

5.2 HC-SR04 Ultrasonic Sensor

The HC-SR04 is used to measure the distance between the sensor and the water surface.

The sensor sends an ultrasonic pulse toward the water surface. The pulse is reflected back to the sensor, and the time taken for the echo to return is measured.

The distance is calculated using:

$$\text{Distance} = (\text{Time} \times \text{Speed of Sound}) / 2$$

The division by 2 is required because the ultrasonic wave travels from the sensor to the water surface and then returns to the sensor.

The measured distance is used to calculate the height of water inside the tank.

5.3 SIM800L GSM Module

The SIM800L is a GSM communication module used to send SMS notifications.

The Arduino communicates with the SIM800L using serial communication. When a critical water-level condition occurs, the Arduino sends an AT command to the GSM module along with the predefined message.

Example low-level SMS:

"ALERT: Water level is LOW. Please refill the tank."

Example full-level SMS:

"ALERT: Water tank is FULL. Please switch OFF the motor."

5.4 Buzzer

The buzzer provides an audible warning when the water level reaches a critical condition.

In this project:

- Below 20% → Buzzer alert

- 20% to below 90% → Normal condition
 - 90% or above → Buzzer alert
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6. CIRCUIT CONNECTIONS

The major connections used in the project are:

HC-SR04 to Arduino Uno

HC-SR04 Pin	Arduino Uno
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VCC	5V
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GND	GND
-----	-----

TRIG	D7
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ECHO	D6
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SIM800L to Arduino Uno

SIM800L Pin	Arduino Uno
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TX	D10
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RX	D11
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GND	GND
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VCC	Appropriate power supply
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The Arduino software uses:

SoftwareSerial gsm(10, 11);

Therefore, Arduino D10 is used as the software serial receive pin and D11 as the software serial transmit pin.

Buzzer to Arduino Uno

Buzzer Pin	Arduino Uno
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Positive (+)	D8
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Negative (-) GND

Note: The SIM800L module requires a suitable stable power supply. Its power requirements should be checked carefully before connecting it to an Arduino supply pin.

7. WORKING PRINCIPLE

The working of the project can be divided into the following stages:

Step 1: Distance Measurement

The HC-SR04 ultrasonic sensor is positioned at the top of the water tank. It sends ultrasonic waves toward the water surface.

The sensor measures the time taken for the reflected wave to return.

Step 2: Water Level Calculation

The Arduino calculates the distance between the sensor and water surface.

The project uses a tank height of:

100 cm

The water height is calculated using:

Water Height = Tank Height – Measured Distance

The water level percentage is then calculated using:

Water Level (%) = (Water Height / Tank Height) × 100

For example, if the water height is 65 cm:

Water Level = (65 / 100) × 100 = 65%

Step 3: Level Classification

The Arduino compares the calculated water level with predefined limits.

LOW LEVEL

If:

Water Level < 20%

The system identifies the condition as LOW.

The buzzer is activated and an SMS is sent to the user.

NORMAL LEVEL

If:

20% ≤ Water Level < 90%

The system identifies the condition as NORMAL.

No warning SMS is sent and the buzzer remains off.

FULL LEVEL

If:

Water Level ≥ 90%

The system identifies the tank as FULL.

The buzzer is activated and an SMS is sent to the user asking them to switch off the motor.

8. SYSTEM FLOW

The overall operation of the system is:

Start

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Initialize Arduino, Ultrasonic Sensor, Buzzer and GSM Module



Measure Distance Using HC-SR04



Calculate Water Height



Calculate Water Level Percentage



Check Water Level



If Water Level < 20%

Activate Buzzer → Send Low-Level SMS



If $20\% \leq$ Water Level < 90%

Normal Condition → No Alert



If Water Level $\geq$ 90%

Activate Buzzer → Send Full-Level SMS



Repeat Monitoring

9. SOFTWARE IMPLEMENTATION

The project is programmed using the **Arduino IDE**.

The main programming functions include:

- Reading the HC-SR04 ultrasonic sensor.
- Calculating distance.
- Calculating water level percentage.
- Comparing the water level with threshold values.
- Controlling the buzzer.
- Communicating with the SIM800L GSM module.
- Sending SMS alerts.

The Arduino program uses the **SoftwareSerial** library to communicate with the GSM module.

The important threshold values used in the program are:

Tank Height = 100 cm

Low Level = Below 20%

Full Level = 90% or above

10. ALERT SYSTEM

The system provides two major alerts.

10.1 Low Water Alert

When the water level falls below 20%, the Arduino activates the buzzer and sends an SMS.

SMS:

"ALERT: Water level is LOW. Please refill the tank."

The system uses an alert flag so that repeated SMS messages are avoided while the tank remains in the same low-level condition.

10.2 Full Tank Alert

When the water level reaches 90% or above, the Arduino activates the buzzer and sends an SMS.

SMS:

"ALERT: Water tank is FULL. Please switch OFF the motor."

The full-level alert is also controlled using a flag to avoid continuously sending the same SMS.

11. ADVANTAGES

The proposed system has several advantages:

1. **Automatic monitoring** – Water level is measured automatically.
 2. **Reduces water wastage** – Full-tank alerts help prevent overflow.
 3. **Remote notification** – SMS alerts allow users to monitor the tank remotely.
 4. **Low-cost implementation** – The system uses easily available components.
 5. **Reduced manual effort** – Users do not need to check the tank frequently.
 6. **Real-time monitoring** – The water level is continuously measured.
 7. **Easy to implement** – The system can be used in homes, schools, and small buildings.
 8. **Expandable system** – Additional features such as automatic motor control can be added.
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12. LIMITATIONS

The system also has some limitations:

1. The HC-SR04 sensor may provide inaccurate readings if the water surface is highly disturbed.
 2. The GSM module requires a suitable mobile network connection.
 3. The SIM800L requires an appropriate and stable power supply.
 4. The system currently provides alerts but does not automatically control the water pump.
 5. Ultrasonic measurements can be affected by sensor positioning and environmental conditions.
 6. The system depends on the availability of GSM service for SMS notifications.
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13. APPLICATIONS

The Smart Water Tank Monitoring System can be used in:

- Residential houses
 - Apartment buildings
 - Schools and colleges
 - Hospitals
 - Offices
 - Hotels
 - Commercial buildings
 - Small industries
 - Agricultural water storage systems
 - Public water storage facilities
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14. FUTURE ENHANCEMENTS

The system can be improved with additional features in the future.

14.1 Automatic Motor Control

A relay module can be added to automatically switch the water pump ON when the water level is low and OFF when the tank becomes full.

14.2 Mobile Application

A mobile application can be developed to display the water level in real time.

14.3 Cloud Monitoring

The system can be connected to an IoT cloud platform to store and visualize historical water-level data.

14.4 Multiple Tank Monitoring

The system can be expanded to monitor multiple water tanks from a single monitoring platform.

14.5 Water Usage Analysis

Historical data can be collected to analyze water consumption patterns and identify unusual usage.

15. RESULT

The developed Smart Water Tank Monitoring System successfully monitors the water level using an HC-SR04 ultrasonic sensor.

The Arduino Uno calculates the water level percentage based on the measured distance. When the water level is below 20%, the system produces a low-level warning and sends an SMS. When the water level reaches 90% or above, the system produces a full-level warning and sends an SMS.

For normal water levels between 20% and below 90%, the system operates without generating an alert.

Therefore, the proposed system provides an effective method for monitoring water tanks and helps reduce water wastage caused by overflow.

16. CONCLUSION

The **Smart Water Tank Monitoring System using IoT** provides a simple and effective solution for automatic water-level monitoring.

The HC-SR04 ultrasonic sensor measures the water level, while the Arduino Uno processes the sensor information and determines the tank condition. The buzzer provides a local warning, and the SIM800L GSM module provides remote SMS notifications.

The system reduces manual monitoring, helps prevent water overflow, and allows users to respond quickly to low or full water-level conditions.

With future improvements such as automatic pump control, cloud connectivity, and mobile applications, the system can become a complete smart water-management solution.

17. REFERENCES

1. Arduino Uno technical documentation.
2. HC-SR04 ultrasonic sensor documentation.
3. SIM800L GSM module documentation.
4. Arduino IDE documentation.
5. SoftwareSerial library documentation.

6. Basic principles of IoT-based monitoring systems.
